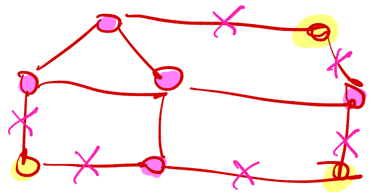
MIN-CUT vs MAX-CUT

poly-time NP-hard

[cut $(S, \bar{S})$ where $\bar{S} = V \setminus S$

and $E(S, \bar{S})$ denotes the cutset



conceptually we find a 2-coloring with maximum number of 2-colored edges

We see three approaches to get a poly-time 2-approximation:

1. local search
2. greedy
3. random coin tossing

1 LOCAL SEARCH

$S = \{\}$

i.e. we switch its color

while $\exists u \in V$ s.t. changing set (i.e. $u \in S \Rightarrow u \in \bar{S}$ or $u \in \bar{S} \Rightarrow u \in S$)

s.t. $|E(S, \bar{S})|$ strictly increases

do if $u \in S$ then $S = S \setminus \{u\}$

else $S = S \cup \{u\}$

① It terminates as $|E(S,\bar{S})|$ can strictly increase at most $|E|$ times

② It provides a 2-approximation (UB $= |E|$ trivially)

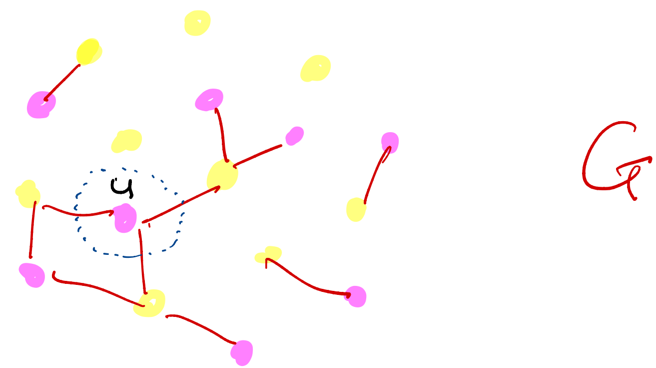
$$\frac{OPT}{\mathrm{Cost}(S)} \leq \frac{UB}{\mathrm{Cost}(S)} \boxed{\leq \frac{|E|}{\mathrm{Cost}(S)} \leq 2}$$

our goal

CLAIM

$$\mathrm{cost}(S) \geq \frac{|E|}{2}$$

cutset $E(S,\bar{S})$



obs $d_u = \mathrm{degree}(u)$

$\forall u \in V$: at least $\frac{d_u}{2}$ incident

(*) edges belong to $E(S,\bar{S})$

(o.w. we can switch color to such a node and strictly increase $|E(S,\bar{S})|$)

$$\mathrm{cost}(S) \overset{\mathrm{def}}{=} |E(S,\bar{S})| = \frac{\sum_{u\in V} \#\text{ edges in } E(S,\bar{S}) \text{ incident to } u}{2} \overset{(*)}{\geq} \frac{\sum_{u \in V} \frac{d_u}{2}}{2} = \frac{\frac{2|E|}{2}}{2} = \frac{|E|}{2}$$

∎

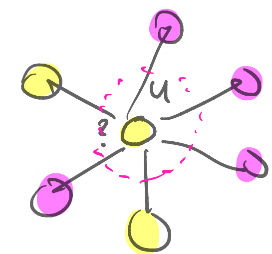
$\boxed{2}$ GREEDY (it works also with weighted graphs in poly-time)

2.a Number the nodes of V from 1 to $n = |V|$ (arbitrary order)

2.b $S = \{\}$

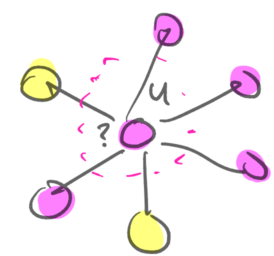
2.c **for** $u = 1, 2, \ldots, n$:

assign u to S or $\bar{S}$ so that $|E(S, \bar{S})|$ is maximized



1) It takes poly-time $O(|V|)$ steps

2) 2-approximation : $\text{cost}(S) \geq \frac{|E|}{2}$



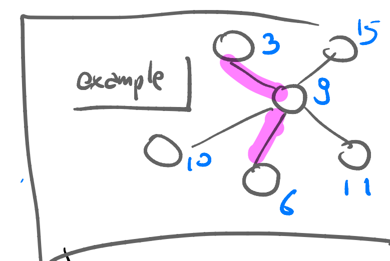
Let's consider edge ij

Suppose w.l.o.g. $i < j$: we observe that the fate of ij is decided by j (i.e. if j chooses or not the same color as i)

Hence, it's j to decide whether $ij \in E(S, \bar{S})$ or not!

j = responsible of edge ij (i.e. the largest between the two end points)

r_u = # edges of which u is responsible



(obs. $r_u \leq d_u$)

$\sum_{u \in V} r_u = |E|$ (as each edge has one responsible)

Analogously to local search, each node u has $\geq \frac{r_u}{2}$ incident edges in $E(S,\bar{S})$ $\quad (*)$

$$\mathrm{cost}(S) \stackrel{\text{def}}{=} |E(S,\bar{S})| = \sum_{u \in V} \#\text{edges in } E(S,\bar{S}) \text{ for which } u \text{ is responsible} \stackrel{(*)}{\geq} \sum_{u \in V} \frac{r_u}{2} = \frac{|E|}{2}$$

now each edge is counted once as it has one responsible

$\blacksquare$

Negative results for approximation

- TSP = travel salesman problem
  - n cities, numbered $1, 2, \ldots, n$
  - D_{ij} = distance/cost to go from city i to city j
  - tour: order of visiting all cities
  - cost(tour) = sum of the distances between consecutive cities

    e.g. tour $= i_1 \, i_2 \ldots i_n \in$ permutation$(1, 2, \ldots, n)$

    $$\text{cost}(\text{tour}) = \sum_{k=1}^{n-1} D_{i_k i_{k+1}} + D_{i_n i_1}$$

    going back to the starting city i_1
  - find the min-cost tour (decision version: cost $\le k$)

Prop. We cannot find a poly-time r-approximation for TSP ($r > 1$)
unless $P = NP$

(i.e. solving TSP approximately is as difficult as solving it exactly)

$\boxed{\text{Idea}}$ GAP method (amplify the weights more than a factor r)

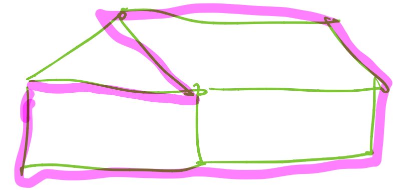
Take HAM (Hamiltonian circuit problem)

$G = (V, E)$ instance of HAM (i.e. we want to know if there is a cycle crossing all nodes exactly once)

By contradiction, suppose A_r is a poly-time r-approx alg. for TSP

A_r poly-time $\Rightarrow$ poly-time

We use A_r to design another algorithm A for $G=(V,E)$



Algorithm A :

1] $D_{ij} = \begin{cases} 1 & \text{if } ij \in E \\ 1 + r \cdot |V| & \text{otherwise} \end{cases}$

2] Execute A_r on the instance D for TSP ← poly-time by contradiction hypothesis

3] Let C be the cost estimated by A_r in step 2]

- <u>if</u> $C \leq r|V|$ <u>then</u> say that G has a Hamiltonian path (YES)
- <u>else</u> say that G is not Hamiltonian (NO)

∎

A takes poly-time

proof: A provides the correct answer to HAM

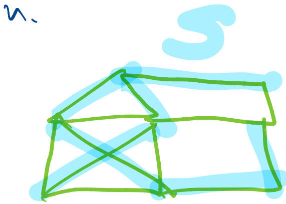
Two cases on G:

- G is Hamiltonian: $\exists$ tour in D such that all $D_{ij} = 1$ as the corresponding edges exist. Hence $OPT = |V|$
  $\Rightarrow C \leq r \cdot OPT = r|V|$ by definition
  $\Rightarrow A$ answers YES

- G is not Hamiltonian: $\forall$ tour in D, at least one $D_{ij} \neq 1$ as the the corresponding edge $ij \notin E$ by construction
  $\Rightarrow OPT \geq \underbrace{(|V|-1) \times 1}_{D_{xy} \geq 1} + \underbrace{(1 + r|V|)}_{D_{ij} \neq 1}$
  $\Rightarrow C \geq OPT \geq (r+1)|V| > r|V| \Rightarrow$
  $\Rightarrow A$ answers NO

$\blacksquare$

Under some restrictions, TSP admits an approximation.



METRIC TSP   Triangle inequalities holds

$$D_{ij} \le D_{ik} + D_{kj}$$

2-approximation (it can be improved)

IDEA See TSP instance as a complete graph $G=(V,E,W)$ whose edges $ij \in E$ have weight $W_{ij} = D_{ij}$ (>0)

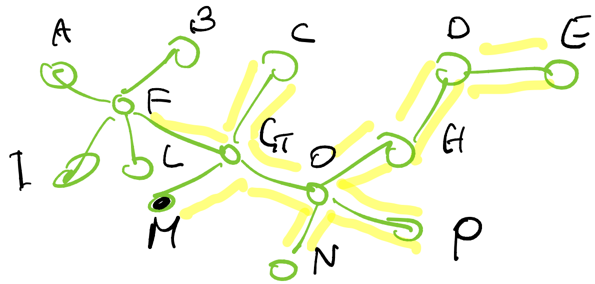
Alg

1] Compute $S = MST(G)$ complete

min-cost spanning tree

2] Compute tour T as follows: perform a pre-order traversal of tree S (from any node) and, whenever a node i is seen for the first time, append i to T

Example

$S =$ 

$T = M\,G\,O\,N\,P\,H\,D\,E\,C\,F \ldots$

obs poly-time alg. as both MST and Traversal are poly-time

Claim cost(T) is a 2-approximation for metric TSP

we need to find a lower bound $LB \le OPT$ to the optimal cost

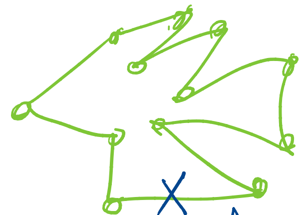
known! → $\dfrac{\text{cost}(T)}{OPT} \le \dfrac{\text{cost}(T)}{LB} \boxed{\le 2}$ what we want to prove...

unknown!

known!

How to find LB :

Any tour without last step (i.e. a cycle traversing all nodes once in which we remove an edge is a spanning tree)



← all nodes in V

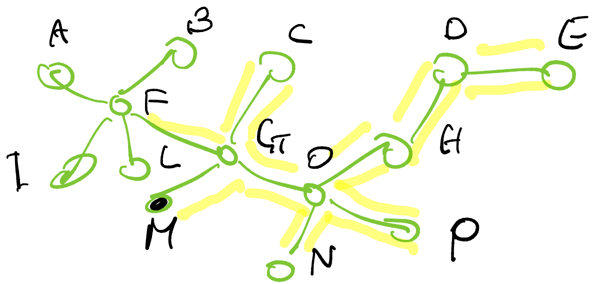
without an edge is a spanning tree

$OPT \geq cost(S)$, $S = MST(G)$ $\Rightarrow$ Hence $LB = cost(S)$

cost of the optimal tour

To show that $\frac{cost(T)}{LB} = \frac{cost(T)}{cost(S)} \leq 2$ using triangle inequality

$S =$



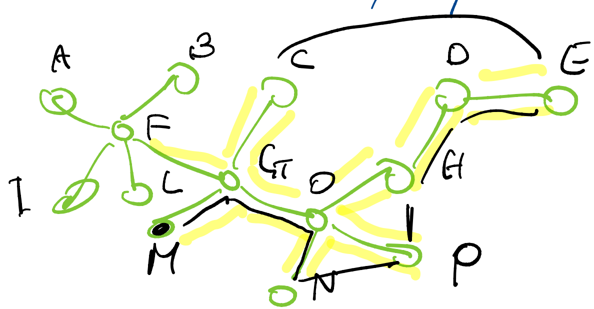
$T = MGONPHDECF\ldots$

If we follow the yellow path, each edge of S is traversed twice

$\Rightarrow$ cost(yellow) $= 2$ cost$(S) = 2\,LB$

Using triangle inequality, we observe that cost$(T) \leq$ cost(yellow) and we are done, by transitivity: cost$(T) \leq 2\,LB$

$S =$



black path corresponds to T (each edge exists as G is complete)